Dynamic computational platform for resource management

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Abstract

Resilience is the vital essence of the smart complex systems, which form the basis of emerging and future services. One of the fundamental challenges in the instantiation of the resilient smart complex systems is the automated allocation of resources. The objective of this research is to develop a dynamic computational platform for supporting long-term systems-level decisions. We simulate a system-level, phenomenological, Markov model of the entire habitat, which enables investigating different scenarios through evaluating the effectiveness of multiple subsets of actions. In the proposed platform, the sub-systems models are input/output models. The intelligent health management is the simplified one that makes a pool of damaged sub-systems and considers the availability of the robotic agents and cost-effectiveness of the recovery, to assign the robotic agents. A new metric is defined to encode value of the system over the design period. The value metric quantifies the initial design cost and recovery action costs while mapping the entire state of the habitat system into mission accomplishment and the crew health.

1 Introduction

Intro: Complexity of a deep space habitat

Gap: Lack of a comprehensive decision support to allow quantitative analysis

Objective: Developing a computational platform to evaluate the long-term consequences of the decisions.

Contribution: Our major contribution is to enable quantification of the interdependecies among sub-systems and capturing the uncertainties originated from hazards, diagnosis and recovery.

Scope:

2 Methodology

2.1 Problem description

Let X_t be the state of the entire habitat (i.e., it is the collection of the states of all subsystems), Y_t be all the relevant sensor data, and A_t be all the relevant actions taken by the health management system policy. Note that there may be other actions taken independently by subsystems, e.g., the humans and the robots have some freedom to act in various ways. It is important to understand that A_t does not include these actions. These actions are internalized. What we have been developing so far with the help of the standard notion is a description of the dynamics of the entire system, i.e., we have dynamics of the form:

$$X_{t+1} = f(X_t, A_t, W_t; d),$$

where W_t is the collection of all external disturbances. Let's d denotes the design decision variable. However, in this work we will assume that it is possible to run this model forward in time. The sensor measurements are given by some noisy function of the state:

$$Y_t = g(X_t, W_t),$$

where, without loss of generality, we are assuming that W_t suffices for describing the sensor uncertainty. Again, the details can be revealed using the standard notation. Here we will assume that we can evaluate this function if we know X_t and W_t.

For the physical sub-system i we define a health state $h_{i,t}$, which is an binary array representing the fault tree of the sub-system.

The union of the health state and subset of the states of the sub-system that have direct effect on the health of the crew or mission accomplishment,$X^j_{i,t}$, are called *Critical states* of the sub-system i and denoted as $C_{i,t} = (h_{i,t}, X^j_{i,t})$. So C_t is the collection of the *Critical states* of the whole system $C_t = (C^1_t, \dots, C^N_t)$.

Health management system. The health management system is doing two jobs:

1. **Fault Detection and Diagnostics(FDD)**: It uses sensor data to identify the hidden state of the system. Specifically, at time t it reports a probability density over the current state of the system: $p_t(C_t|y_{1:t-1})$.
2. **Real-time Decision Making (DM)**: It uses sensor data to decide what to do next. The action is $A_t = \mu_t(p(C_t|y_{1:t-1})$.

So, a health management system is completely described by:

$$\mathcal{H} = ((p_1, \mu_1), (p_2, \mu_2), \dots). \tag{1}$$

Given a health management system, the system *critical states* transition probability become:

$$p(C_{t+1}|C_t, A_t = \mu_t(p(h_{i,t}|y_{1:t-1}), W_t; d). \tag{2}$$

2.2 Transition probability decomposition

Considering the N number of sub-systems, we may need to decompose this transition probability in this fashion:

$$p(C_{t+1}|C_t, W_t, A_t, d) = \prod_{i=1}^{N} p(C^i_{t+1}|\mathcal{C}^i_t) \tag{3}$$

where $\mathcal{C}^i_t$ denote the set of sub-system states that share an interdependency with sub-system i, with the following form

$$\mathcal{C}^i_t = \{C^j_t, 1 \leq j \leq N\}. \tag{4}$$

It is important to clarify that C^j_t is in subset $\mathcal{C}^i_t$ if and only if sub-system j shares an interdependency with sub-system i.

2.3 Objective Performance Metrics

Objective performance metrics are functions of the trajectory of the *Critical states* of the system up to a given time. Note that we are using the “state” and **not** the sensor measurements because the sensor measurements may be wrong. Also, we are using the “state” and **not** the “identified state,” because the identified state may be also wrong. For the sake of simplicity, let us assume that we are working on a finite time horizon T. A performance metric is just a function of the *Critical states* trajectory, the actions up to time T, and the design parameters. So, it is:

$$\Pi_T = \pi(C_{1:T}, A_{1:T}; d). \tag{5}$$

We take the expectation and variance of the performance metric over all possible trajectories. We define a generic performance metric as a function of the *Critical states* trajectory and the actions up to time T plus design parameters:

1. **Value metric:** Here the value of the entire system over the design period T is denoted as V_T. Assume that being at a given state C_t at time t has value $v_z(C_t)$, that picking design parameters d has a specific cost $l(d)$, and that picking a specific action has also a given cost $l_a(A_t)$. Then, we can define the *net present value metric* by:

$$V_T = \sum_{t=1}^{T} e^{-\beta t} \left(v_z(C_t) - l_a(A_t)\right) - l(d), \tag{6}$$

where β is a discount factor. To unify the units for measuring the value and the cost, Let's assume that we are working in mass equivalent, Then:

   (a) The term $l(d)$ is the initial cost in mass equivalent of a specific design.

   (b) The term $l_a(A_t)$ is the cost of a performing a given action. Maybe the action costs material that should be provided form earth. Maybe the action destroys a spare part or a robot. Maybe the action destroys a robot (very high cost). Maybe the action kills a human (very high cost). The action costs can probably be to zero (assuming that we do not include in the possible actions catastrophic ones). Cost of energy consumption can be tricky. If we consider a situation that we may happen to skip an action because of high energy consumption then we should consider this cost. All costs should be converted to mass equivalent.

   (c) The term $v_z(C_t)$ can be decomposed further using the state zones. This can become arbitrarily complicated. To simplify the problem to a solvable scale, we assume that the $v_z(C_t)$ can takes values associated to the two dimensional value zones. The value zones are describing the *accomplishment of the mission* and *Health of the crew* given the state of the system (C_t), e.g., green (value zone 1), a yellow (value zone 2), a orange (value zone 3), and a red (value zone 4), see Fig. 1. Being in the green zone gives us some value per unit of time. Being in the yellow costs a bit per unit of time. Being in the orange costs considerable amount per unit of time. Being in the red, costs a lot per unit of time. We can write this as:

$$v_z(C_t) = \sum_{C_{1,t},\dots,C_{N,t}} v_{C_{1,t},\dots,C_{N,t}} \prod_{i=1}^{N} 1_{Z_{C_{i,t}}}(C_t^i), \tag{7}$$

where $v_{S_{1,t},\dots,S_{N,t}}$ represent the value of experiencing the set of systems states of $S_{1,t},\dots,S_{N,t}$ for one unit of time (negative if there is an associated cost). $Z_{C_{i,t}}$ is the zone set, i.e., a subset of the state space, and $1_{Z_{C_{i,t}}}$ is the characteristic function of the set $Z_{C_{i,t}}$. Defining the $Z_{C_{i,t}}$ requires the domain knowledge of the specific sub-system.

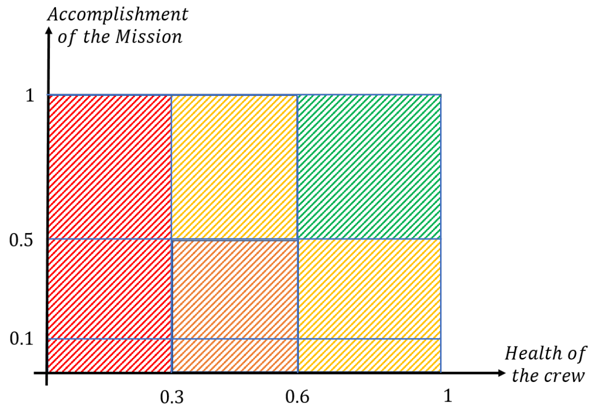


Figure 1: Schematic of the cost function.

3 Illustrative Example

As demonstrated in the Figure 1, the thermal management case study consists of multiple sub-systems listed as below: 1. Power generation 2. Power storage 3. Robotic agents 4. Structural mechanical 5. Structural thermal 6. Thermal unit 7. Interior Environment 8. Human agents (not shown in figure to make it less complicated) 9. Intelligent health management (IHM)

Disturbances to the system are defined as all the external excitations from the outside of the system. The interdependencies between the subsystems are defined as below:

- Interdependency: Two sub-systems are interdependent if the state of one of them depends on the state other. In this way, disturbances in one sub-system can ripple through to other subsystems.

- Physical interdependency: Two sub-systems are physically interdependent if they are interdependent and the link is physical. Examples of physical links include transfer of heat, transfer of mechanical energy, transfer of power, transfer of mass.

- Cyber interdependency: Two sub-systems are cyber interdependent if they are interdependent and the link is informational. For example, the thermal sub-system sends a temperature measurement to the intelligent health management (IHM) sub-system and the IHM sub-system may change the temperature set-point of the thermal sub-system.

- Repair agents: Special sub-systems that receive informational input from the IHM, potentially physical input from a storage sub-system, and can provide physical inputs to other sub-systems. Robots and humans are both repair agents. An example of the action of a robot agent is the following. The IHM detects that dust has accumulated on the PV panels and that repair is needed. Then the IHM sends a signal to a robot agent and assigns it the task of cleaning the PV panels (information passing from IHM to robot). The robot agent moves to the storage facility, picks up a “broom” tool (mass transfer from storage to robot), goes to the PV panel and cleans it (physical input from robot to PV panel).

- Physical intervention: A physical intervention represents the physical coupling of repair agents with a given sub-system. From the previous example, the robot picking up a “broom” tool from the storage facility is a physical intervention. Similarly, the robot cleaning the PV panel is a physical intervention.

- Cyber intervention: Cyber intervention represents the commands from intelligent health management (IHM) to the sub-systems. Consequently, the states of the sub-systems depend on outputs of the IHM. The outputs of the IHM are inputs to the other sub-systems, which are passed

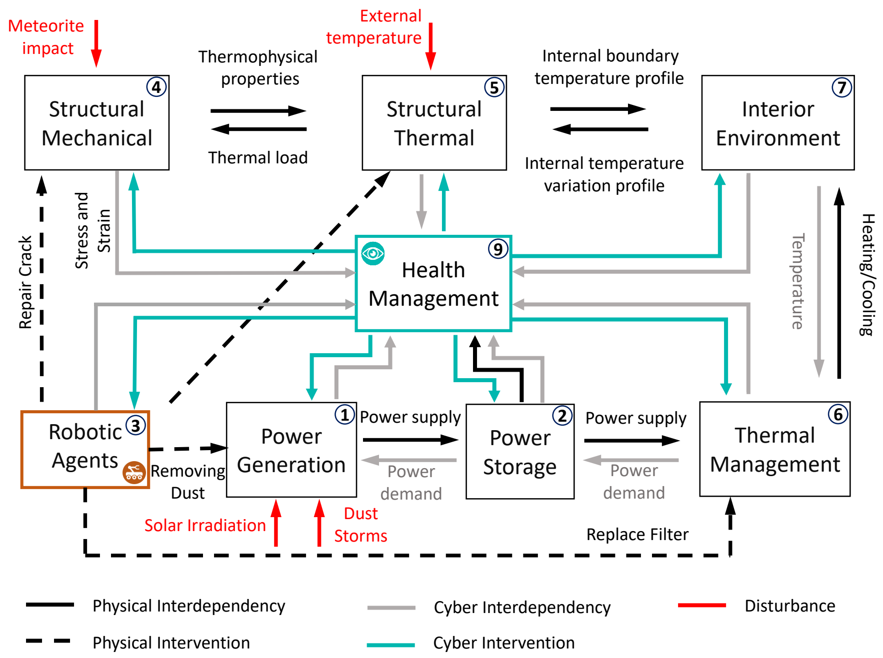


Figure 2: The schematic of the thermal management system. The figure does not show human agents (whose role is similar to the robotic agents). Also, not all possible disturbances or interventions are shown.

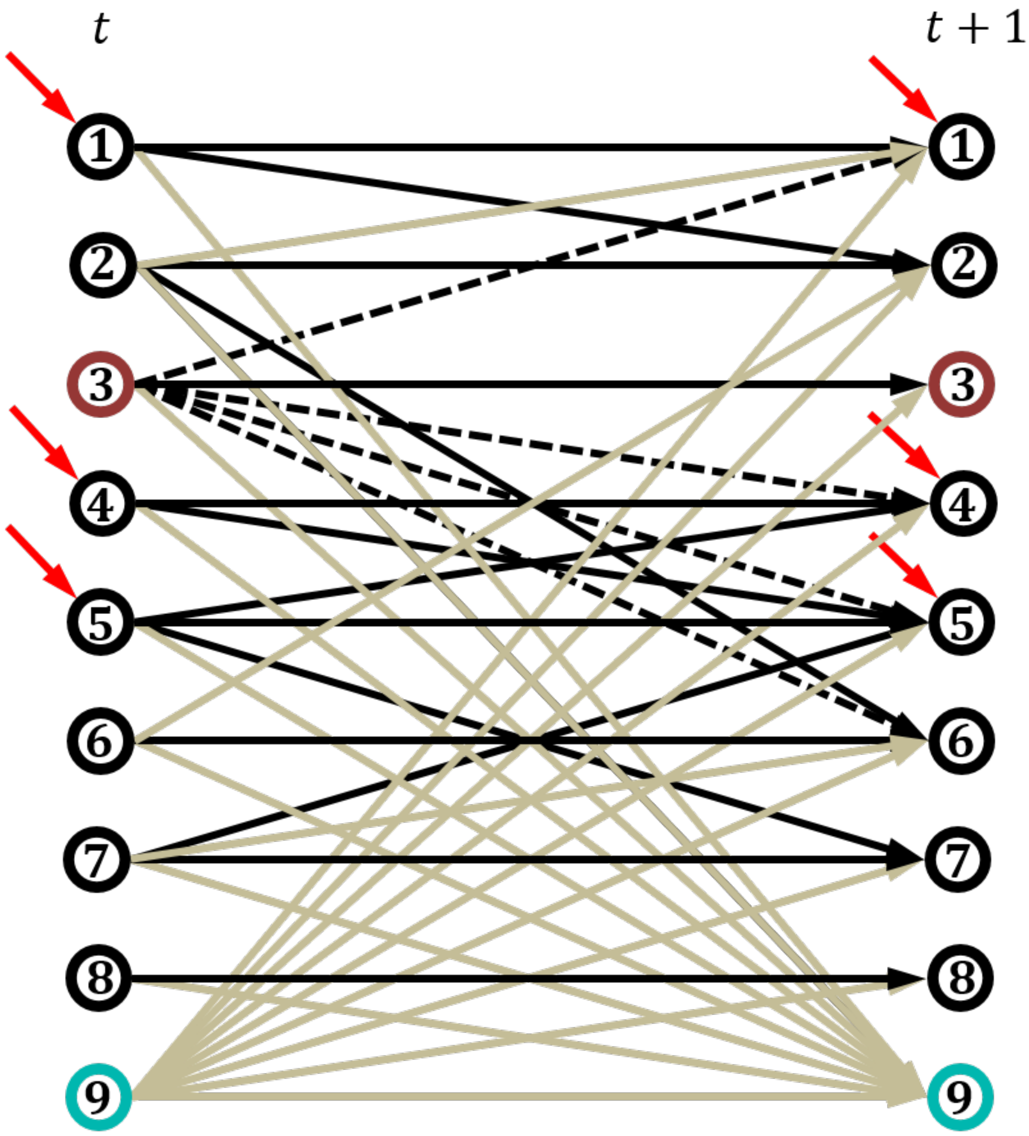
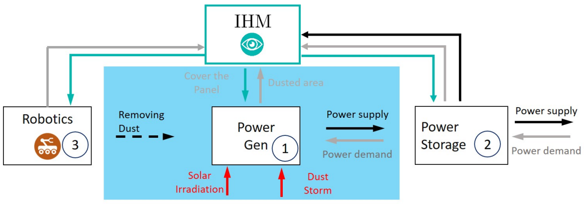
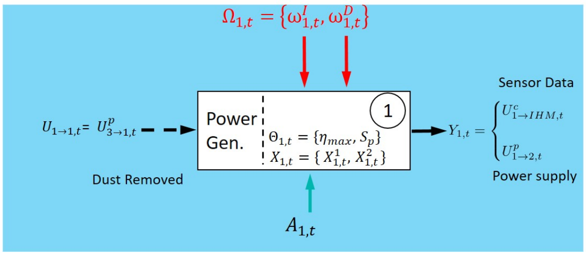


Figure 3: Dynamic Baysian Network representation of the system demonstrated in Figure 2



(a) Schematic of the power generation sub-system and the surrounding sub-systems.



(b) The power generation sub-system in standard notation.

Figure 4: Schematic of the system.

in the form of information.

3.1 Power generation sub-system

Here we focus on describing an oversimplified model of the power generation subsystem, shown in Figure 2. Figure 2(a) shows the relationship between the power generation sub-system and the surrounding sub-systems. Figure 2(b) demonstrates the power generation sub-system in the form of standard notation.

$X_{1,t} = (X^1_{1,t}, X^2_{1,t})$: The state of the system at time t (2D vector).

$X^1_{1,t}$: The amount of dust covering the PV panel area (real number in [0,1], unitless). Zero (0) means that the panels are clean. One (1) means that the panels are fully covered and that no energy can be produced. The assumption is that any number between zero and one corresponds to an intermediate state that reduces energy efficiency proportionally (check assumption).

$X^2_{1,t}$: A binary variable capturing whether the PV panel is covered (0) or open (1) (unitless). If the panel is covered, then no power is produced but also not dust is being accumulated. If the panel is closed, power is produced during the day, but dust is being accumulated. The panel can be closed during the night since there is no available solar irradiance.

$U^p_{3\to1,t}$: A possible physical intervention of the robotic agents to the power generation sub-system representing the percentage of the PV panel area cleaned by the robotic agents in one unit of time.

$\theta_1 = \eta, S_p$: The set of physical (design) parameters of the power generation unit:

η : the nominal efficiency of the PV cells S_p: the total surface area of the PV cells.

$A_{1,t}$: A signal sent from the IHM sub-system to the power generation sub-subsystem. Here we have only possibility: a binary signal in which zero (0) means do nothing and one (1) means flip the state cover state of the panel (from covered to open and from open to covered).

$\Omega_{1,t} = (\omega^I_{1,t}, \omega^D_{1,t})$: Stochastic vector process representing external disturbances. Specifically:

$\omega^I_{1,t}$: Solar irradiation, the amount of electromagnetic radiation received from the sun per unit area, (W/m^2).

$\omega^D_{1,t}$: The amount of dust being deposited on the PV panel area in a unit of time (nonnegative real number, unitless). This variable increases significantly when there is a dust storm.

$Y_{1,t} = (Y^P_{1,t}, Y^C_{1,t}, Y^I_{1,t})$: The output of the power generation sub-system. The first component $Y^P_{1,t}$ is the generated power (W). The second component $Y^C_{1,t} = X^2_{1,t}$ is the covered/open binary variable (which can be estimated using a sensor). The third component $Y^I_{1,t} = \omega^I_{1,t}$ is a the solar irradiance (which be measured using a pyranometer).

$U^p_{1\to2,t} = h_{12,t}(Y_{1,t}) = Y^P_{1,t}$: The power supply to the power storage unit that represent the physical coupling between the output of power generation unit and the power storage, (W).

$U^c_{1\to IHM,t} = m_{1,IHM,t}(Y_{1,t}) = (Y^P_{1,t}, Y^C_{1,t}, Y^I_{1,t})$: The information passed to the IHM. Note that we have included $Y^C_{1,t}$ and $Y^I_{1,t}$ because they are sufficient for the IHM to estimate the current dust accumulation $X^1_{1,t}$.

3.1.1 Dynamics of the power generation unit state

The amount of the dust at time $t+1$ is equal to the amount of the dust at time t plus the amount of dust accumulated gradually and in case of a dust storm when the power generation system is open subtracted by the amount of dust has been cleaned by the robotic agents. This value is bounded below by 0 and above by 1. The dynamics are as follows:

$$X_{1,t+1} = min\{(X^1_{1,t} + \omega^D_{1,t} X^2_{1,t} - U^p_{3\to 1,t})_+, 1\} \tag{8}$$

where we have used the shorthand notation:

$$(x)_+ = \begin{cases} x & \text{if } x \geq 0 \\ 0 & \text{if } x < 0 \end{cases} \tag{9}$$

The dynamics of the cover/open binary variable depend on the signal send from the IHM system. This signal simply flips the state. This can be modeled as:

$$X^2_{1,t+1} = X^2_{1,t} + A_{1,t}(mod2), \tag{10}$$

where (mod 2) stands for addition modulo 2, which is simply the result of the addition followed by taking the remainder the division by 2. For example, 0 + 0 (mod 2) = 0, 0 + 1 (mod 2) = 1, 1 + 0 (mod 2) = 1, 1 + 1 (mod 2) = 0. For the last one, 1 + 1 is 2 divided by 2 gives remainder zero.

The power produced is:

$$Y^P_{1,t} = g_1(X_{1,t}, \theta_1, \omega_{1,t}) = (1 - X^1_{1,t}) X^2_{1,t} \omega^I_{1,t} \eta S_p. \tag{11}$$

3.2 Robotic sub-system

3.3 Simplified Health management

3.4 Transition Probability Decomposition

* Temporal text: Based on the system shown in Fig. 2 and interdependencies shown in the Bayesian network representation in Fig. 3, the transitional probability decomposition in Eq. 21 can be derived.

First, the probabilities for each sub-system and interdependencies subset $\mathcal{C}^i_t$ and described in Eq. 4 are derived.

<u>Sub-System 1:</u> Power Generation

$$p(C^1_{t+1} | C^1_t, C^2_t, C^3_t, C^9_t) \tag{12}$$

<u>Sub-System 2:</u> Power Storage

$$p(C^2_{t+1} | C^1_t, C^2_t, C^6_t, C^9_t) \tag{13}$$

<u>Sub-System 3:</u> Robotic Agents

$$p(C^3_{t+1} | C^3_t, C^9_t) \tag{14}$$

<u>Sub-System 4:</u> Structural Mechanical

$$p(C^4_{t+1} | C^3_t, C^4_t, C^5_t, C^9_t) \tag{15}$$

<u>Sub-System 5:</u> Structural Thermal

$$p(C^5_{t+1} | C^3_t, C^4_t, C^5_t, C^7_t, C^9_t) \tag{16}$$

<u>Sub-System 6:</u> Thermal Management

$$p(C^6_{t+1} | C^2_t, C^3_t, C^6_t, C^7_t, C^9_t) \tag{17}$$

<u>Sub-System 7:</u> Interior Environment

$$p(C^7_{t+1} | C^5_t, C^7_t, C^9_t) \tag{18}$$

Sub-System 8: ? Not in diagram

$$p(C_{t+1}^{8}|C_{t}^{8}, C_{t}^{9}) \tag{19}$$

Sub-System 9: Health Management

$$p(C_{t+1}^{9}|C_{t}^{1}, C_{t}^{2}, C_{t}^{3}, C_{t}^{4}, C_{t}^{5}, C_{t}^{6}, C_{t}^{7}, C_{t}^{8}, C_{t}^{9}) \tag{20}$$

Then, the probability for state of the system, becomes

$$p(C_{t+1}|C_t, W_t, A_t, d) = \prod_{i=1}^{9} p(C_{t+1}^{i}|\mathcal{C}_{t}^{i}) \tag{21}$$

4 Results

5 Conclusion